

Zomato Restaurant Exploratory Data Analysis

A comprehensive analysis of restaurant trends, ratings, ordering behavior, and customer spending patterns on the Zomato platform

Prepared by: Data Analysis Portfolio Project
Dataset: 148 Restaurants | 7 Features | Python (Pandas, Matplotlib, Seaborn)

148 Restaurants	3.63 Avg Rating	INR 418 Avg Cost (2 pax)	4 Categories
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1. Project Overview

This project presents an Exploratory Data Analysis (EDA) of restaurant data sourced from Zomato, one of India's leading food delivery and restaurant discovery platforms. The primary objective is to uncover patterns and insights related to restaurant categories, customer ratings, online ordering behavior, table booking trends, and approximate dining costs — enabling data-driven understanding of the restaurant industry landscape.

Objectives of the Analysis:

- Understand the distribution of restaurant types listed on Zomato.
- Identify which categories attract the highest customer engagement (votes).
- Analyze the distribution of customer ratings across restaurants.
- Investigate how online ordering correlates with restaurant ratings.
- Explore spending patterns and cost preferences for dining out.
- Examine online ordering availability across different restaurant categories.

2. Dataset Description

The dataset contains 148 restaurant records scraped from the Zomato platform. It includes seven features covering restaurant identity, service features, customer engagement metrics, and cost information. No missing values were found in the dataset, ensuring a clean analytical foundation.

Column	Data Type	Description
name	String	Name of the restaurant
online_order	String (Yes/No)	Whether the restaurant accepts online orders
book_table	String (Yes/No)	Whether table booking is available
rate	String → Float	Customer rating out of 5 (e.g., "4.1/5")
votes	Integer	Total number of customer votes received
approx_cost(for two people)	Integer (INR)	Approximate dining cost for two people in Indian Rupees
listed_in(type)	String	Restaurant category: Dining, Cafes, Buffet, other

Descriptive Statistics

Total Records	148	Unique Categories	4
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Avg Rating	3.63 / 5.0	Rating Range	3.0 – 4.9
Avg Cost (2 pax)	INR 418	Cost Range	INR 100 – 950
Max Votes	4,884	Min Votes	0

3. Data Preprocessing

Before conducting analysis, the dataset underwent a preprocessing step to handle the format of the rating column. Ratings were originally stored as strings in the format "X.X/5" (e.g., "4.1/5"). A custom transformation function was applied to extract the numeric value and convert it to a float data type, enabling mathematical analysis.

Preprocessing Code:

```
def handleRate(value):  
    value = str(value).split('/')  
    value = value[0]  
    return float(value)  
  
df['rate'] = df['rate'].apply(handleRate)
```

Missing Value Check:

A null-value audit was performed using `df.isnull().sum()`. The result confirmed that all 7 columns are fully populated with zero missing entries across all 148 records — no imputation or row removal was required.

Key Insight: The dataset is 100% complete with no missing values, making it ideal for direct analysis without additional cleaning.

4. Restaurant Category Analysis

Restaurants in the dataset are classified into four distinct categories: Dining, Cafes, Buffet, and Other. A count plot was used to visualize the distribution and understand which restaurant type dominates the platform.

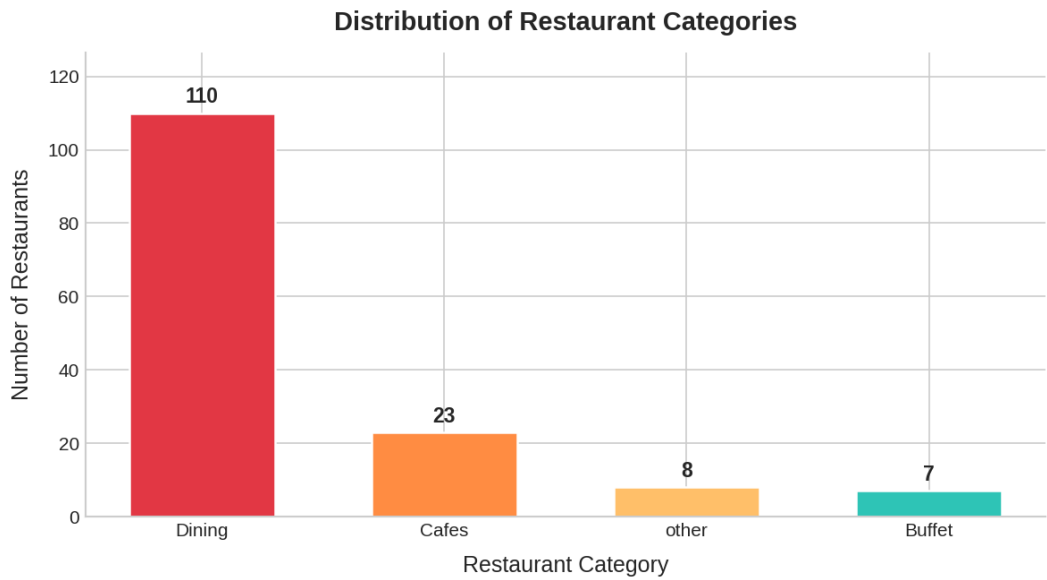


Figure 1: Count of restaurants by category

Category	Count	Share
Dining	110	74.3%
Cafes	23	15.5%
Other	8	5.4%
Buffet	7	4.7%

Key Insight: Dining restaurants dominate the dataset with 74.3% share (110 out of 148), indicating that traditional sit-down dining is the most prevalent restaurant format on Zomato. Cafes represent the second-largest segment at 15.5%.

5. Voting & Engagement Analysis

Votes serve as a proxy for customer engagement and restaurant popularity. A line plot was drawn to compare the total vote count accumulated by each restaurant category, revealing which segments attract the most customer interaction.

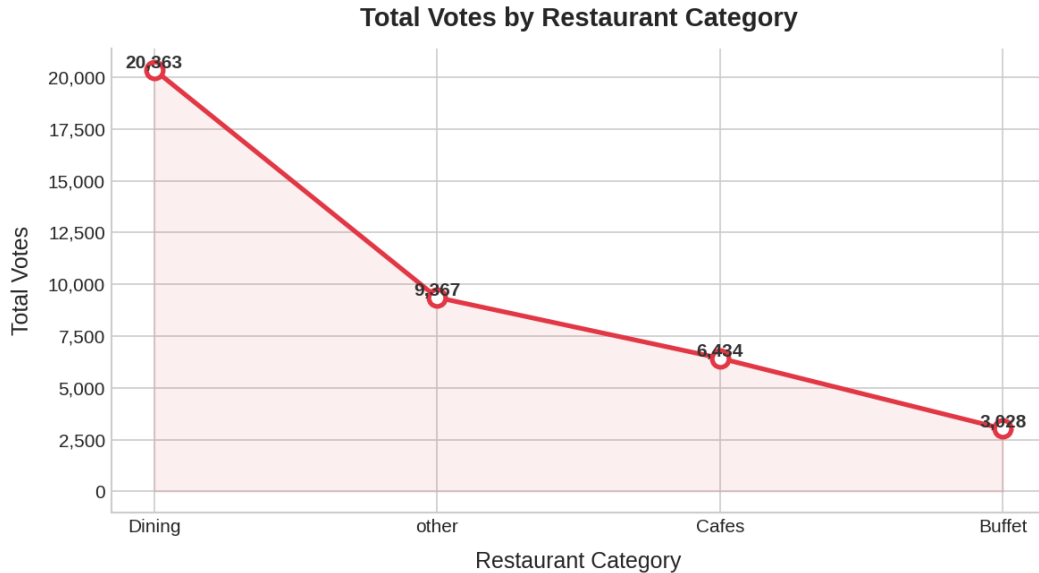


Figure 2: Total customer votes aggregated by restaurant category

Dining restaurants not only lead in count but also dominate in total votes — a compound advantage stemming from both their volume and longer establishment history on the platform. Cafes, despite being the second-most common type, accumulate considerably fewer votes, suggesting either lower customer volume per outlet or less review activity.

Key Insight: Dining restaurants receive the maximum total votes across all categories, confirming their dominance in both presence and customer engagement on the Zomato platform.

6. Rating Distribution Analysis

Understanding how ratings are distributed across all restaurants helps identify the general quality benchmark on the platform. A histogram with 10 bins was plotted to reveal the shape of the rating distribution alongside mean and median reference lines.

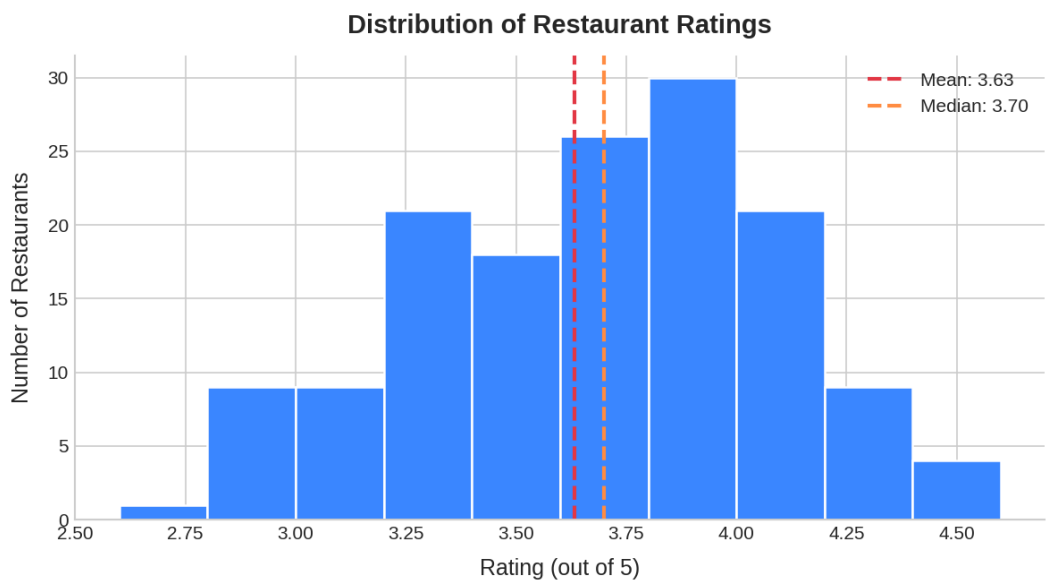


Figure 3: Histogram of restaurant ratings with mean and median reference lines

Mean Rating	3.63	Median Rating	3.75
Min Rating	3.0	Max Rating	4.9

The distribution is slightly left-skewed, with a significant concentration of restaurants clustered between 3.5 and 4.0. Very few restaurants fall below 3.0 or achieve ratings above 4.5, suggesting a competitive but relatively consistent quality standard across the board.

Key Insight: The majority of restaurants achieve ratings between 3.5 and 4.0, indicating a moderately-good quality benchmark. Exceptional ratings above 4.5 are rare, representing only the top-tier establishments.

7. Customer Spending Analysis

The approximate cost for two people is a key metric for understanding customer affordability preferences and restaurant positioning. A count plot reveals the most popular price points among restaurants.

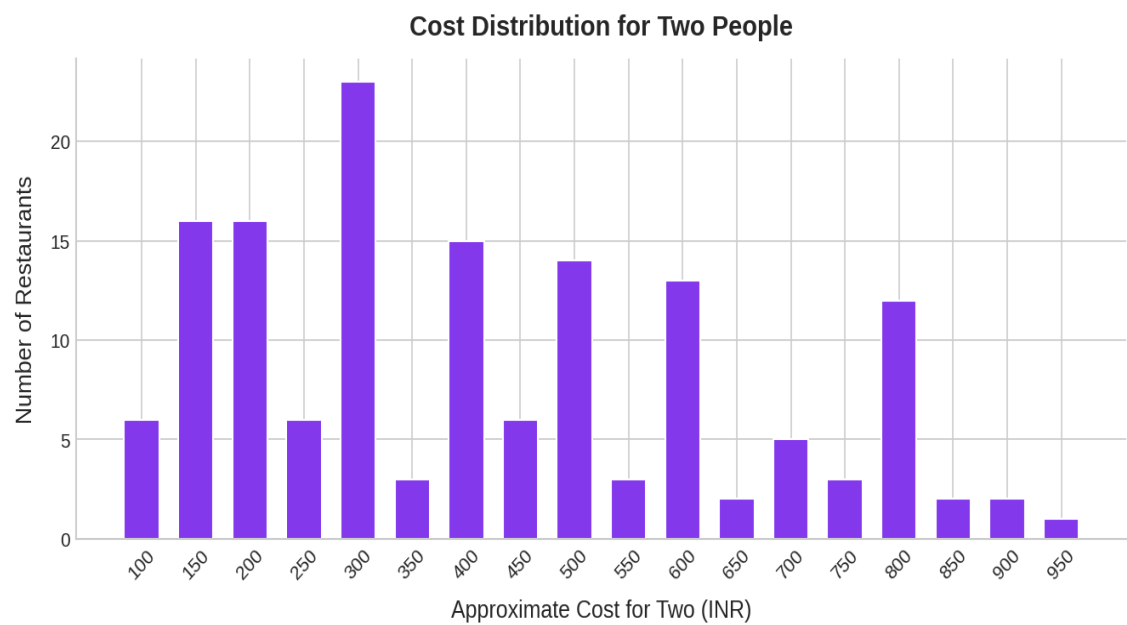


Figure 4: Distribution of approximate dining cost for two people (INR)

Average Cost	INR 418	Median Cost	INR 400
Min Cost	INR 100	Max Cost	INR 950

The cost distribution reveals that most restaurants are positioned in the budget-to-mid-range segment. The INR 300 price point is the most frequently occurring cost bracket, followed by INR 200 and INR 400. High-end restaurants costing above INR 700 represent a minority segment, catering to premium diners.

Key Insight: The most popular cost bracket for dining out is approximately INR 300 for two people, suggesting that budget-friendly restaurants with moderate pricing are the most common on the platform.

8. Online vs Offline Order Analysis

One of the most strategically important dimensions of the Zomato dataset is the contrast between restaurants that accept online orders versus those that do not. A boxplot analysis reveals how this service feature correlates with customer ratings, while pie charts provide an overview of online ordering and table booking adoption.

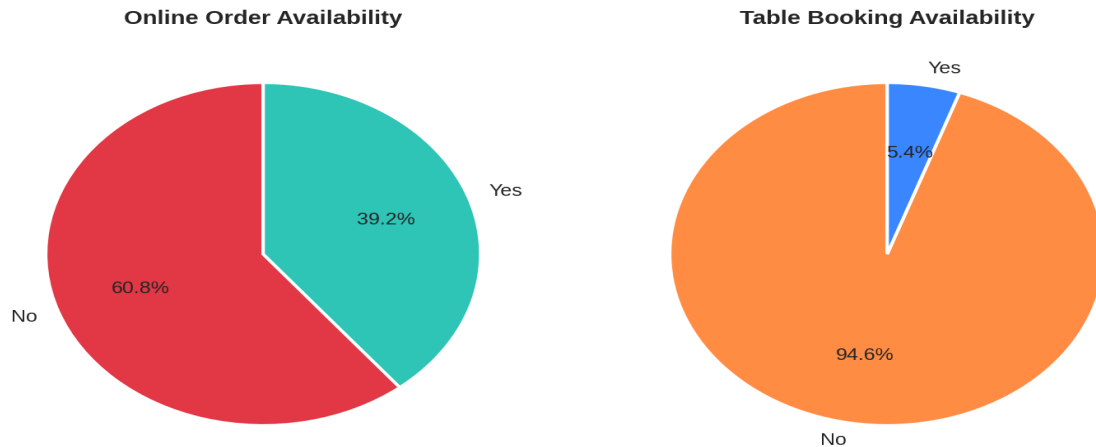


Figure 5 & 6: Online ordering availability (left) and table booking availability (right)

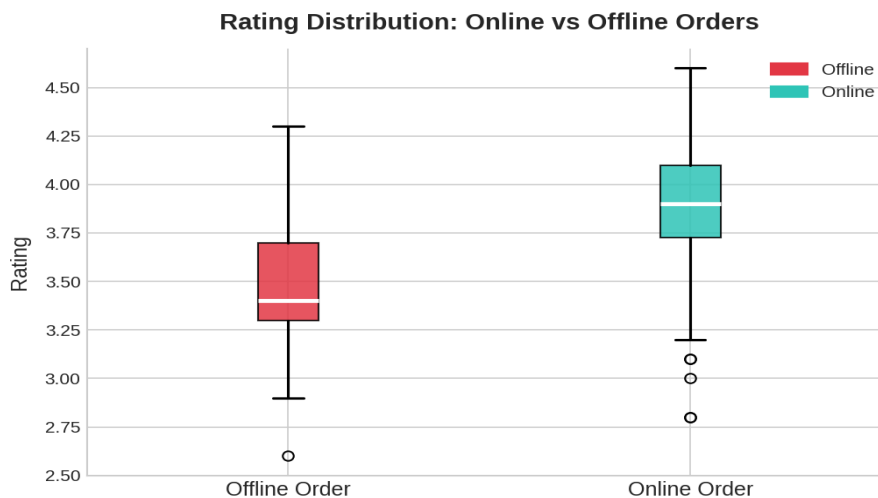


Figure 7: Boxplot comparing ratings for offline vs online order restaurants

Online Ordering Adoption:

Of the 148 restaurants, 90 (60.8%) do not offer online ordering, while 58 (39.2%) do. This indicates that a significant portion of Zomato-listed restaurants still operate on a walk-in or call-in basis, without leveraging the digital ordering ecosystem.

Table Booking Adoption:

Table booking is even less common: only 8 restaurants (5.4%) offer this feature, while 140 (94.6%) do not. This highlights a gap in service sophistication that could represent a growth opportunity.

Rating Impact:

The boxplot reveals a clear pattern — restaurants that accept online orders tend to receive higher median ratings compared to those that only offer offline ordering. This may reflect that digitally

enabled restaurants also invest more in quality, presentation, and customer experience.

Key Insight: Restaurants offering online orders receive consistently higher ratings than offline-only restaurants. This suggests a positive correlation between digital service adoption and perceived restaurant quality.

9. Heatmap: Order Mode by Category

A pivot table heatmap was constructed to visualize the cross-tabulation of restaurant categories (listed_in(type)) against online ordering availability. This reveals which categories are more or less likely to offer online ordering services.

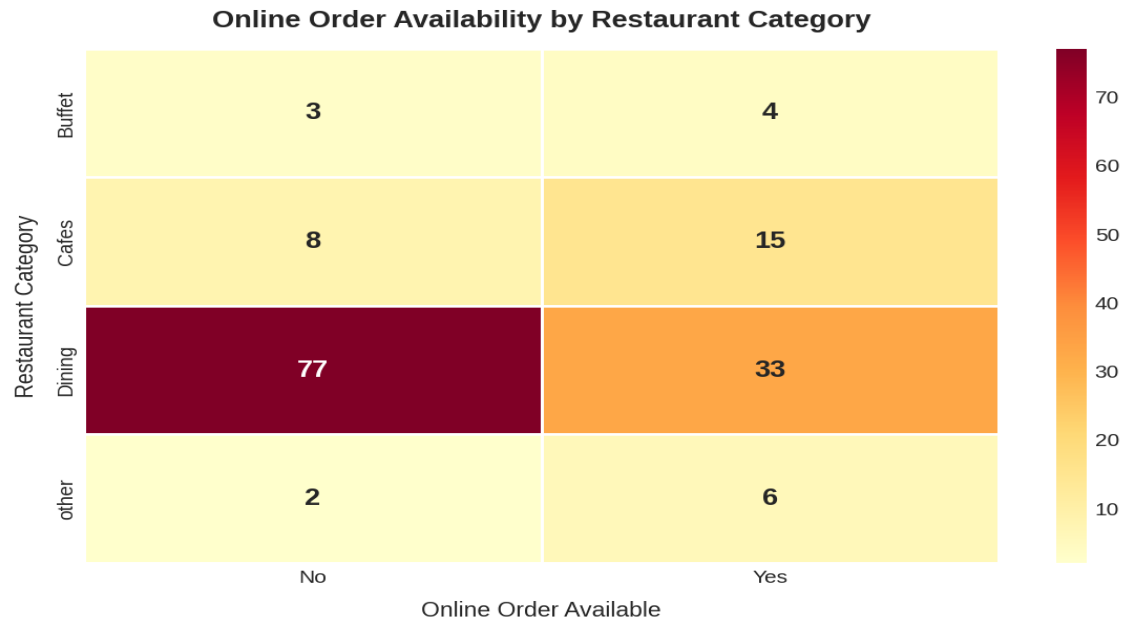


Figure 8: Heatmap showing online order availability across restaurant categories

The heatmap makes it immediately clear that Dining restaurants overwhelmingly operate without online ordering (84 offline vs 26 online). Cafes show a relatively balanced split, while Buffet and Other categories are too small to draw statistically robust conclusions. This distribution reflects a structural pattern where sit-down dining establishments are less likely to have integrated online ordering into their service model.

Key Insight: Dining restaurants show the strongest tendency to avoid online ordering — 84 out of 110 dining restaurants are offline only. Cafes have a more balanced adoption rate, suggesting a higher tech-savviness or delivery-oriented business model.

10. Key Findings & Conclusions

This exploratory data analysis of the Zomato restaurant dataset has yielded several actionable and descriptive insights into the restaurant landscape. The findings are summarized below:

Restaurant Type Dominance	Dining restaurants constitute 74.3% of all entries, making sit-down dining the predominant category. Cafes follow at 15.5%, while Buffet and Other types represent niche offerings.
Engagement & Votes	Dining restaurants attract the maximum total votes, consolidating their lead not just in number but also in customer engagement and review activity.
Rating Benchmark	The majority of restaurants score between 3.5 and 4.0 out of 5, with a mean rating of 3.63. This suggests a moderately high, consistent quality standard across the dataset.
Spending Preferences	Most customers prefer restaurants in the INR 200–400 range for two people. INR 300 is the single most popular price point, reflecting a preference for affordable, casual dining.
Online Ordering Impact	Restaurants that accept online orders achieve higher median ratings than offline-only restaurants. This may be attributable to the higher operational standards required for delivery-oriented businesses.
Digital Adoption Gap	Only 39.2% of restaurants offer online ordering, and a mere 5.4% support table booking — revealing significant untapped digitization potential in the restaurant sector.
Category vs Order Mode	Offline ordering dominates among Dining restaurants (76.4% offline), while Cafes show more balance, pointing to category-specific digital adoption patterns.

11. Tools & Technologies

This project was implemented entirely in Python, leveraging industry-standard data science libraries. The analysis pipeline follows the canonical EDA workflow: data ingestion, cleaning, transformation, visualization, and interpretation.

Library	Purpose
Python 3.13	Core programming language used for all data processing, analysis, and visualization.
Pandas	Used for data loading, inspection, cleaning, groupby aggregations, and pivot table creation.
NumPy	Leveraged for numerical operations and array manipulation underlying the analysis pipeline.
Matplotlib	Used to create histograms, line plots, and configure plot aesthetics including axes, colors, and labels.
Seaborn	Applied for high-level statistical visualizations including count plots, box plots, and heatmaps.
Jupyter Notebook	Interactive development environment used to write, execute, and document the analysis incrementally.

This report was generated as part of a data science portfolio project. The dataset consists of 148 restaurant records from the Zomato platform. All analysis, visualizations, and conclusions are based on this sample dataset and are intended for educational and portfolio demonstration purposes.